

Bridging stratigraphic palaeobiology and phylogenetics

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Organizers

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Abstract

Stratigraphic paleobiology and phylogenetics seek to understand how physical and biological processes (respectively) shape the fossil record. Yet, these fields are poorly integrated. The stratigraphic record is the key source of temporal information used to constrain the evolution of life and is the purview of Earth scientists. Phylogenetics, although it relies on this temporal record to reconstruct evolutionary time trees, allowing us to fill critical gaps in the fossil record, is largely the purview of biologists. Few studies exploit both approaches simultaneously. The challenge of reconciling the physical geological record with phylogenetics lies in the theoretical chasms that exist between disciplines, as well as limitations created by computational tools, technical skills, and ontologies of uncertainty. Meanwhile recent advances in modelling in both areas create opportunities for joint models of stratigraphic and phylogenetic processes. In this workshop, we will isolate these opportunities and unite stratigraphers, evolutionary biologists and software engineers to solve them together interactively. This workshop aims to establish a workflow for incorporating the geological record into phylogenetic inference, to create a roadmap for the digital infrastructure required for successful integration of geological and evolutionary data, and to translate these theoretical advances into actionable research software improvements.

Scientific case

Phylogenetics is the science of reconstructing evolutionary relationships, rates and processes. It is built through a concerted effort of mathematicians, palaeobiologists, geneticists, and computer scientists. In the last decades, genetic sequences from living and – in exceptional circumstances – recently extinct taxa have become the primary source of data for phylogenetic trees. Such data represents a tip-of-an-iceberg proportion of known diversity, as over 99% of organisms are extinct and either discoverable only as **fossils** or lost for good. For many reasons, which can be argued mathematically and are understood intuitively, the inclusion of fossils in phylogenetic trees, regardless of the methodological approach, radically improves our understanding of evolution (Ronquist et al. (2012), Gavryushkina et al. (2017); Saladin et al. (2017)).

Recent mathematical and computational developments, collectively known as the **Fossilized Birth-Death** (FBD) family of models, treat fossils and extant species as observations from the same macroevolutionary process, rather than external calibration constraints, providing statistically principled methods for integrating temporal, morphological, and molecular evidence. The introduction of the FBD process has led to major revisions in evolutionary timelines across diverse groups from insects to mammals, demonstrating that fossil

data contains irreplaceable information about deep-time evolutionary processes, reviewed in (Mulvey, Nikolic, et al. (2025)).

Among the features of the FBD model that allow it to incorporate fossil data is the fact that it is a **diversification and sampling model**, meaning (1) that the inference of the topology can directly incorporate information about fossil ages, and (2) that we can make explicit assumptions about incomplete sampling and the structure of available fossil data (Stadler 2010). These aspects of the FBD model, the sampling process and fossil ages, constitute the interface between phylogenetics and geosciences. The FBD model has enabled us to incorporate stratigraphic information directly into phylogenetic inference. For example, fossils can virtually never be dated directly and the typical precision of a fossil age is several million years. Ignoring stratigraphic age uncertainty can introduce errors of 10-20 million years in divergence time estimates but this source of uncertainty can readily be accounted for in a Bayesian framework (Barido-Sottani et al. 2020). Beyond this, much more untapped potential remains. Other information about the stratigraphic record, such as variation in sampling through time has almost never been accounted for in analyses using FBD models.

Stratigraphic paleobiology is the discipline studying the laws and processes governing fossil preservation and discoverability. It offers a theoretical framework and a predictive toolbox for identifying gaps in the fossil record, environmental factors affecting fossil occurrences, and quantification of the uncertainty of fossil ages (Patzkowsky and Holland 2012). In this workshop, we aim to address the challenges of using fossil data in phylogenies using a stratigraphic palaeobiology approach.

Research challenges of the field

Interdisciplinarity

Tapping into the record of evolution in rocks requires, essentially, bridging the divide between natural and physical sciences. A molecular phylogeneticist can rapidly familiarise themselves with genetic data for a new group. In contrast, fossils are almost exclusively known from their morphological remains, often fragmentary, which require expert knowledge of anatomy, preservation and geological sampling to translate into data useful for phylogenetics. Yet, meta-analyses of combined molecular-morphological datasets demonstrate that morphological and molecular information on the same sets of organisms frequently show conflicting signals when analyzed separately, but combined analyses often yield unique trees not sampled by either type of information (partition) individually (Keating, Garwood, and Sansom 2023). Even when morphological data has a lower information contents on its own, it consistently improves phylogenetic resolution and accuracy when combined with molecular data (Asher and Smith 2022). We want to address the interdisciplinary data-model integration problem: What empirical data can realistically be used to inform phylogenetic tree inference from fossils and stratigraphy, on which timescales, and how to obtain this data?

Understanding the geological record

The geological record offers a conceptual challenge: it can be represented in the stratigraphic (depth) domain and in the time domain. The stratigraphic domain is the physical record: the strata, the fossils, the boundaries and event horizons, and the minerals bearing radioisotopes that allow calibration in absolute time. The time domain is where the strata and fossils are formed, where extinction and origination happen, and what we cannot observe but must reconstruct from the stratigraphic domain. Most information needed for phylogenetics is implicitly expressed in the time domain (fossil ages, rates, sampling), but taken from the stratigraphic domain. The stratigraphic record is commonly treated as a continuous, lossless mapping from the time domain, but the broad and diverse fields of stratigraphy and sedimentology provide a wealth of processes that challenge this approach. Recent advances in modeling and geochronology (led by researchers at Utrecht University) allow us to build representations of time in geological strata, including their uncertainty (Hohmann et al. 2024), but these tools are yet to be incorporated in phylogenetic inference.

Models of morphological evolution

Beyond providing temporal information, fossils also preserve morphological characteristics in their appearance, which constitute an important source of data for phylogenetic inference. The use of discretized morphological

data in phylogenetics has traditionally relied on parsimony-based methods. More recently, however, their application within a Bayesian framework has gained traction, offering the advantage of directly incorporating fossil data into time-calibrated phylogenetic inference while jointly resolving relationships among both extinct and extant lineages (Wright and Hillis 2014). Nevertheless, the number of morphological models available to describe character change remains limited, reflecting the difficulty of generalizing trait evolution. The most widely used approach is the Mk model, which assumes equal transition rates among states and equal stationary frequencies (Lewis 2001). Several extensions of this model have been developed and are frequently applied, although their assumptions about character evolution remain relatively simplistic. Despite these limitations, studies indicate that such models perform well and are capable of capturing essential aspects of the evolutionary process (Mulvey, May, et al. 2025). An important aspect of fossil character data that remains underexplored is the influence of stratigraphic and taphonomic biases, both on the fossils themselves and on the loss of entire specimens from the fossil record. Gaining a better understanding of how these biases shape our observed data could inform strategies to model the processes that generated it, integrating insights from both biological and geological perspectives.

Digital infrastructure

A major barrier to integrating information from the fossil record into phylogenetic inference is the fragmentary nature of computational resources used in different fields. Genetic, morphological, and fossil age data are each housed in different databases. Users of phylogenetic software encounter major barriers in software performance. Analyses with stratigraphic constraints require excessive computation power, making the method inaccessible to many researchers with limited resources and IT knowledge. Professional software development processes and governance are needed to support researchers so that they can focus on contributing expert knowledge into relevant packages and using them for research questions. Current paucity of such support is a threat to science and careers: (junior) researchers may spend many months on calculations that may prove faulty and even lead to retractions of publications. Integrating available information therefore requires cross-disciplinary knowledge of data collection practices and advanced statistical methods. By incorporating research software engineers (RSE) in this workshop, we aim to inform researchers on ways to professionalize research software development and to co-create a roadmap for digital infrastructure, which can serve as the basis for well-argued, targeted grant applications.

Aims of the workshop

Our goal is to achieve a framework for the integration of phylogenetics and stratigraphic palaeobiology, to communicate this to the scientific community through a publication, and foster collaboration between scientists contributing to both fields and research software engineers essential to making this research possible.

Benefit to the Dutch research community

The Netherlands has a strong tradition in evolutionary biology, but its share and impact of publications in natural sciences has been falling in the recent years, compared to other EU and world countries, according to the Rathenau Instituut. At the same time, the Netherlands has established itself as a leader in Open Science and software sustainability and sovereignty, through the work of the Netherlands eScience Center and the integration of engineering and research expertise in research strategies and funding programs. Through this workshop, we want to leverage this leadership position. We will promote software tools developed in the Netherlands and integrate this unique expertise into the workflows and tools that are fundamental to phylogenetics and stratigraphic paleobiology. The workshop involves a strong international participation as a way to offer Dutch researchers the opportunity to access the most exciting developments in the field and to counteract increasing scientific isolationism visible in Europe and worldwide. Our goal is to strengthen the role of Dutch scientists in international collaboration through publications, joint strategies and roadmaps, and co-creation of grant applications and projects.

Workshop outcomes

Short-term impact

At the end of the workshop, we will have produced:

1. A **best-practices recommendations list** for publishing and reviewing phylogenetic analyses and reporting fossil and stratigraphic information.
2. A **roadmap for digital infrastructure** needed for landmark use-cases of phylogenetic analysis incorporating fossil data. It will contain a list of priorities from the user perspective.
3. A **technical draft of a manual for phylogenetic reconstruction from simulated data**, addressing uncertainties, sources of error, and instructions for practitioners - this draft will become the basis for a future publication and tutorial.

We will also have **trained** and **built capacity** in using methods of the respective disciplines among phylogeneticists interested in incorporating geological knowledge and stratigraphic palaeobiologists aiming to bring in their expertise into phylogenetics. As research software engineers are still relatively new roles in the academic landscape, we will have trained participants in the fundamentals of communicating with them.

Long-term impact

1. A chapter on using fossil and stratigraphic data in phylogenetic analyses and on replicability in a new planned article on best practices.
2. Translation of researcher needs into digital infrastructure: based on the roadmap for digital infrastructure, feature requests including specifications of missing parts of algorithms and documentation will be prepared, following best practices in research software design. This will allow matching software capabilities with scientific needs and will ensure proper validation of the complex computations involved in phylogenetic inference. The roadmap will serve as a basis for targeted funding applications.
3. A scientific article on phylogenetic inference from incomplete (gappy) fossil data and on incorporating stratigraphic precision and accuracy.

Program

Each day will be dedicated to one problem, to reduce cognitive load, and will be led by one or two of the organizers.

Monday: the structure of the geological record

Emilia Jarochowska and Johannes Hidding

Lorentz Center welcome and opening

Introductory lecture by Mark Patzkowsky: **Principles of Stratigraphic Palaeobiology**

Ice-breaker group coding activity: Make your own geological record. In a pre-workshop survey we will ask participants about their coding skills and divide them into “novice”, “familiar” and “expert”. In the workshop, we will make sure that novice users are always paired with experts and that senior researchers are paired with ECRs. In this workshop, participants will learn the basics of creating a simple model of the geological record on their own computers, using CarboKitten developed jointly at Utrecht University and Netherlands eScience Center (Hidding et al. 2025). This exercise has been tested in previous workshops and is suitable for a diverse audience.

In-depth lecture by Lydian Boschman: **Palaeogeography and plate tectonics in evolution**

Diving deep group coding activity: Extracting information for phylogenies from strata. We will use the models created in the morning to simulate information which can be used in phylogenetic analyses:

age-depth models, fossil ranges including their uncertainties, apparent origination and extinction rates. These datasets will be used in the following days.

Welcome reception

Tuesday: Models of morphological evolution

Laura Mulvey

Introductory lecture by April Wright and Sergei Tarasov **Modelling morphological evolution**

An overview of the conceptual foundations of morphological evolution, including trait coding, evolutionary models, and their application in phylogenetic inference.

Group coding practical: Incorporating morphological data into phylogenetic analysis: In this workshop, participants will gain hands-on experience running phylogenetic analyses using morphological data. They will be tasked with selecting appropriate models and providing justification for their choices. The results of the analyses will be discussed, highlighting challenges encountered and exploring potential approaches for improvement.

Potential Lecture: **Extensions to Common Morphological Models** This session could cover recent methodological developments and extensions to standard models of morphological evolution.

Breakout group activity: Challenges in morphological modeling In this workshop, participants will be grouped together into break-out rooms to discuss practical and theoretical challenges in morphological modeling. Topics may include, deciding which traits are informative for phylogenetic inference, how taphonomic biases influence morphological character evolution, and the advantages and limitations of using discrete versus continuous traits. The discussion will aim to generate insights into developing modeling approaches that better reflect the biological and geological processes shaping the observed data.

Wednesday: Tree inference from fossil data

Niklas Hohmann and Rachel Warnock

Introductory lectures by Philip Donoghue: **Fossil Evidence in Time Tree Reconstruction** and Alexandra Gavryuskina: **Introduction to the Fossilized Birth-Death Model.**

These talks will introduce the major modelling frameworks used to infer time constrained trees, along with conceptual challenges.

Group coding practical: A simulation-based approach to integrating stratigraphic paleobiology and phylogenetics. This session will bring together aspects from all previous workshops, with a focus on forward modelling. Participants will learn how to bring together simulation tools that can be used to simulate stratigraphy, trees, fossil and morphology. This practical session aims to help participants develop an intuition for the model parameters. In addition, we will discuss the different use cases for simulations, including validating and comparing different implementations of the FBD model and exploring limitations of existing models.

Lecture by Rachel Warnock: **Extensions to the FBD processes and challenges**

Breakout group activity: The stratigraphic record as a source of information: Using their knowledge from the workshops on Monday about the data the stratigraphic record provides, participants will discuss in small groups how to characterize information from the stratigraphic record for time tree inference, how analyses from existing studies could be refined and identify novel research questions that can be addressed using more precise information from the stratigraphic record. In addition, we will identify deficits in the methodological landscape and available empirical data sets, i.e., cases of model-misspecification and current limits to including stratigraphic data, and will discuss best practices and strategies for reporting to increase reproducibility of phylogenetic inferences.

Evening: Workshop dinner

Thursday: Best practices and replicability in Bayesian phylogenetics using the fossil record

Rachel Warnock and Niklas Hohmann

Introductory lecture by Ugne Stolz: **Community led phylogenetic software development**

Breakout group activity: Preparing and reporting data for phylogenetic inference. In contrast to previous activities, which focus more on conceptual aspects of extracting information from the stratigraphic record for use in phylogenetics, this day will focus on more practical aspects. This will include an introduction to Bayesian phylogenetic software (BEAST2, MrBayes, RevBayes) and participants will walk through the decision-making process that goes into setting up an analysis using the FBD model. One of the objectives will be to develop best practices for reporting on this decision-making process and ensuring reproducibility. We will also discuss the challenges associated with comparing model implementations and output across different software.

Lecture by Joëlle Barido-Sottani: **Architecture and limitations of common frameworks used in phylogenetic software development.**

Breakout group activity: Digital infrastructure for tree inference from fossil data. Remixing the groups from the morning workshop, we will group together researchers from different disciplines in break-out rooms to establish the major conceptual and infrastructural barriers to incorporating fossil data into tree inference. These will serve as the basis for the *roadmap for digital infrastructure*. Ideally, an RSE will be assigned to each group to balance user and developer perspectives.

Friday: Interactive synthesis

All organizers

We will dedicate the day to carrying out **hands-on phylogenetic analyses** representing two common scenarios: tree inference from incomplete geological record using the FBD Skyline model and tree inference accounting for fossil occurrence uncertainty using the FBD Range model. We will use simulated data of which we have perfect knowledge and the goal will be to recover the original tree properties.

We will apply the *literate programming* approach: documenting our work in human language alongside code. The created document will be the basis for a scientific publication, as well as a tutorial for other users. We will again make sure to form small groups of members with different expertise, i.e., self-declared experts with novices. The produced code will be shared and open and will be the basis for a future tutorial and article reporting the results of the workshop.

Due to high computation time of the analyses they may not be finished by the end of the workshop, although we will provide SURF computing infrastructure to ensure as rapid completion as possible. We will finish the day and the workshop with a joint revision of what has been achieved. To this goal, we will offer the ECRs the opportunity to present a summary of the reconstruction and the results (if already available) on behalf of their teams and to be the ambassadors of the respective team's concerns, recommendations, and findings.

Finally, we will identify a suite of research questions that can be addressed in future simulation and empirical studies, using the tools introduced throughout the week.

Participants

To achieve the aims of the workshop, we need to bring together participants representing: phylogenetics, stratigraphy, bioinformatics, evolutionary biology, sedimentary modelling, and research software engineering. We aim at:

- workshop participants with at least elementary experience with Bayesian phylogenetics, even if they are at the beginning of a PhD journey,
- a small number of key participants representing stratigraphic palaeobiology,

- Research Software Engineers in an advisory role.

Since digital infrastructure is among the challenges in this field, we target particularly participants with at least an intention to engage with the software or research software engineers and to discuss the interplay between the needs of the users and the computational possibilities. The engagement does not mean experience with coding. It also means promoting the enforcement of good software governance and reproducibility among journal editors, supervisors, grant reviewers and other people in senior academic roles.

Since the workshop will include hands-on sessions, including writing or running simple code (with ample help and instruction from the organizers), we intend to survey the participants about their skills and assign them to groups with mixed background, so they can work on problems together.

We aim at a minimum proportion of ECRs of ca. 40% and anticipate between 40 and 50 participants.

Dates

1. 21-25 September 2026
2. 28 September - 2 October 2026
3. 14-18 September 2026

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